FACULTY OF INFORMATION TECHNOLOGY CTU IN PRAGUE

Assignment of bachelor's thesis

Title:	Interfaces for TNL library data structures and algorithms in Python and Julia
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Supervisor:	doc. Ing. Tomáš Oberhuber, Ph.D.
Study program:	Informatics
Branch / specialization:	Software Engineering 2021
Department:	Department of Software Engineering
Validity:	until the end of summer semester 2026/2027

Instructions

1. Become familiar with the basics of the TNL library, in particular its main algorithms and data structures.
2. Become familiar with PyTNL, the Python interface to the TNL library.
3. Explore options for writing lambda functions for computations in TNL using Python. Propose a suitable solution. Alternatively, propose a different approach that enables users to efficiently modify TNL data structures directly from Python.
4. Establish the foundations of the jTNL project, i.e., an interface to the basic data structures of the TNL library for the Julia language.
5. Explore suitable options for writing lambda functions for computations in TNL using Julia, or alternatively propose another approach for efficiently modifying TNL data structures directly from Julia.
6. Apply the acquired knowledge to create interfaces for selected algorithms and data structures from TNL in Python and Julia, e.g., connecting GPU solvers for linear programming problems in TNL to CVXPY or JuMP, or providing interfaces to the TNL-SPH module for Julia and Python.

Electronically approved by Ing. Michal Valenta, Ph.D. on 9 February 2026 in Prague.